

To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic

Code:

```
geya@LenovoL480: ~/OS/Skil x + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];
    char variable[50];
    char *value;

    printf("Enter variable reference (example: $HOME): ");
    scanf("%s", input);

    if (input[0] == '$') {
        strcpy(variable, input + 1);

        value = getenv(variable);

        if (value != NULL) {
            printf("Variable: %s\n", variable);
            printf("Value: %s\n", value);
        } else {
            printf("Undefined variable\n");
        }
    } else {
        printf("Not a variable reference\n");
    }

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 4$ ./task1
Enter variable reference (example: $HOME): $HOME
Variable: HOME
Value: /home/geya
geya@LenovoL480:~/OS/Skill/Week 4$ ./task1
Enter variable reference (example: $HOME): $A
Undefined variable
geya@LenovoL480:~/OS/Skill/Week 4$ |
```